

Medical physics

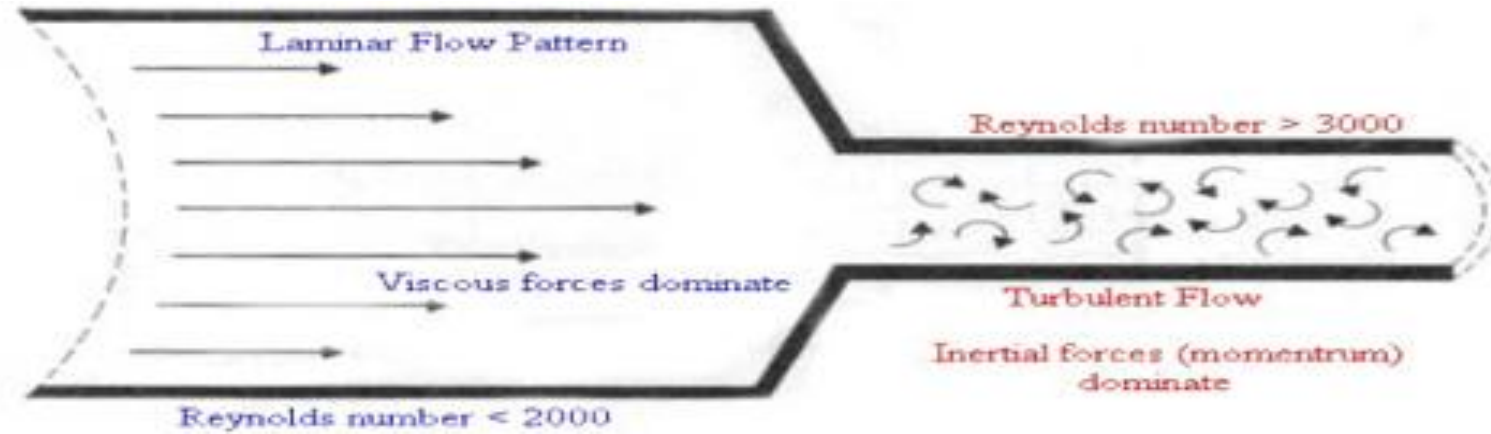
SGL4



Physics of Cardiovascular system Part II

Blood flow-Laminar and Turbulent

- An important characteristic of laminar flow is that it is silent, if all blood flow were laminar, information could not be obtained from the heart with a stethoscope
- The heart sounds heard with a stethoscope are caused by turbulent flow



- If you gradually increase the velocity of fluid flowing in a tube by reducing the radius of the tube, it will reach a critical velocity V_c when laminar flow changes into turbulent
- $V_c = k\eta/\rho r$
- η : viscosity
- R : radius
- ρ : density
- K : reynolds number=1000 for many fluids including blood, flowing in tubes of constant diameter. If there are bends the reynolds number becomes much smaller

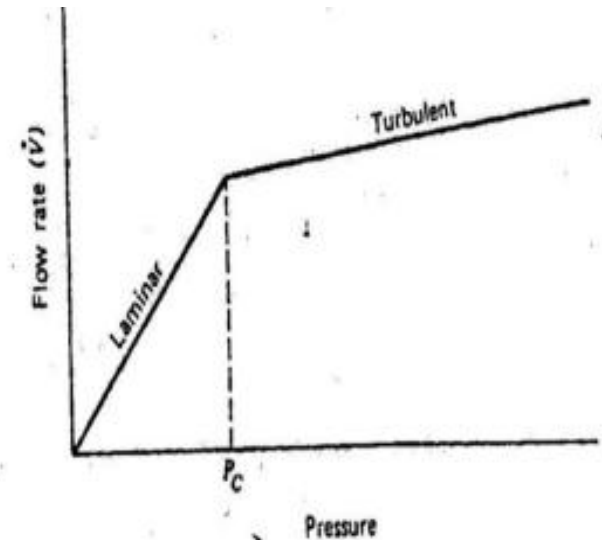
- K: reynolds number to determine whether the fluid flow is laminar or turbulent.
- The critical velocity will be lower if there are restrictions or obstructions in the tube. Reynolds determined that the critical velocity is proportional to the viscosity η of the fluid and is inversely proportional to the density ρ of the fluid and the radius R of the tube, $V_c = K\eta/\rho R$.

In aorta , which has a radius of about 1cm in adult

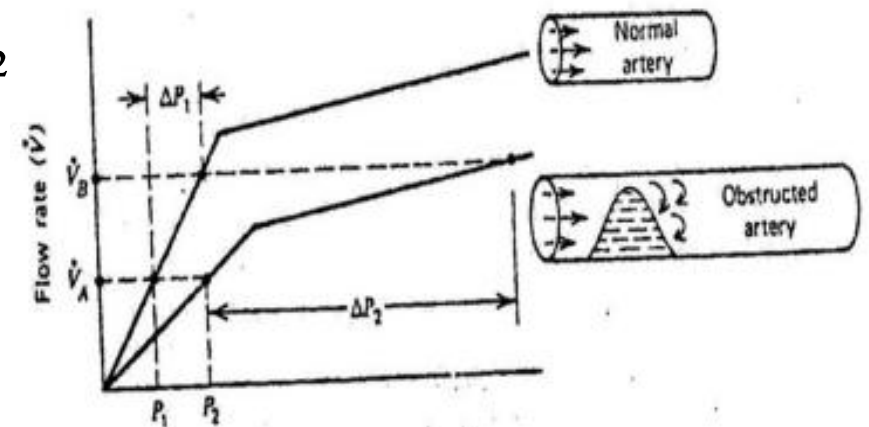
$$\text{The } V_c = K\eta/\rho R = \frac{1000 \times (4 \times 10^{-3} \text{ pas})}{(10^3 \text{ kg/m}^3) (10^{-2} \text{ m})} = 0.4 \text{ m/sec}$$

- The velocity in the aorta range from 0 to 0.5 m/s, and thus the flow is turbulent during part of the systole

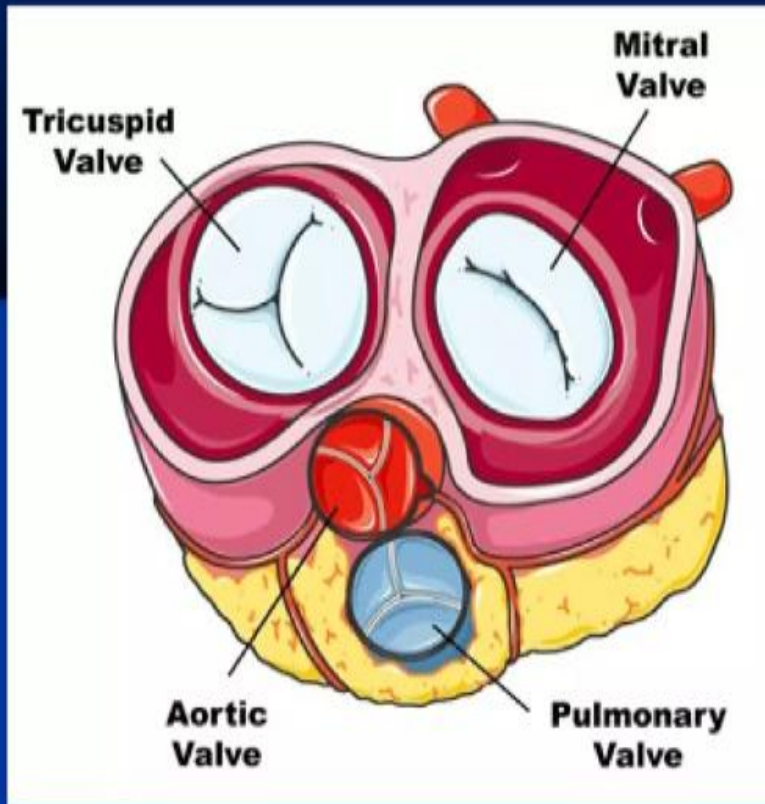
- Laminar flow is more efficient than turbulent flow. This is illustrated graphically in figure a. The slope of the curve in the laminar flow region is greater than that in the turbulent flow region. That is, a given increase in pressure causes a greater increase in the laminar flow rate than in the turbulent flow rate. The reduction in efficiency is apparent in the blood flow through an artery with obstruction (Figure b). For the flow rate V_A a pressure of P_1 is needed for the normal artery and some what higher pressure P_2 is needed for the obstructed artery. If both arteries are required to deliver a new flow rate V_B , the increase in pressure ΔP_2 will be much greater for obstructed artery since the flow will be turbulent.



(a)

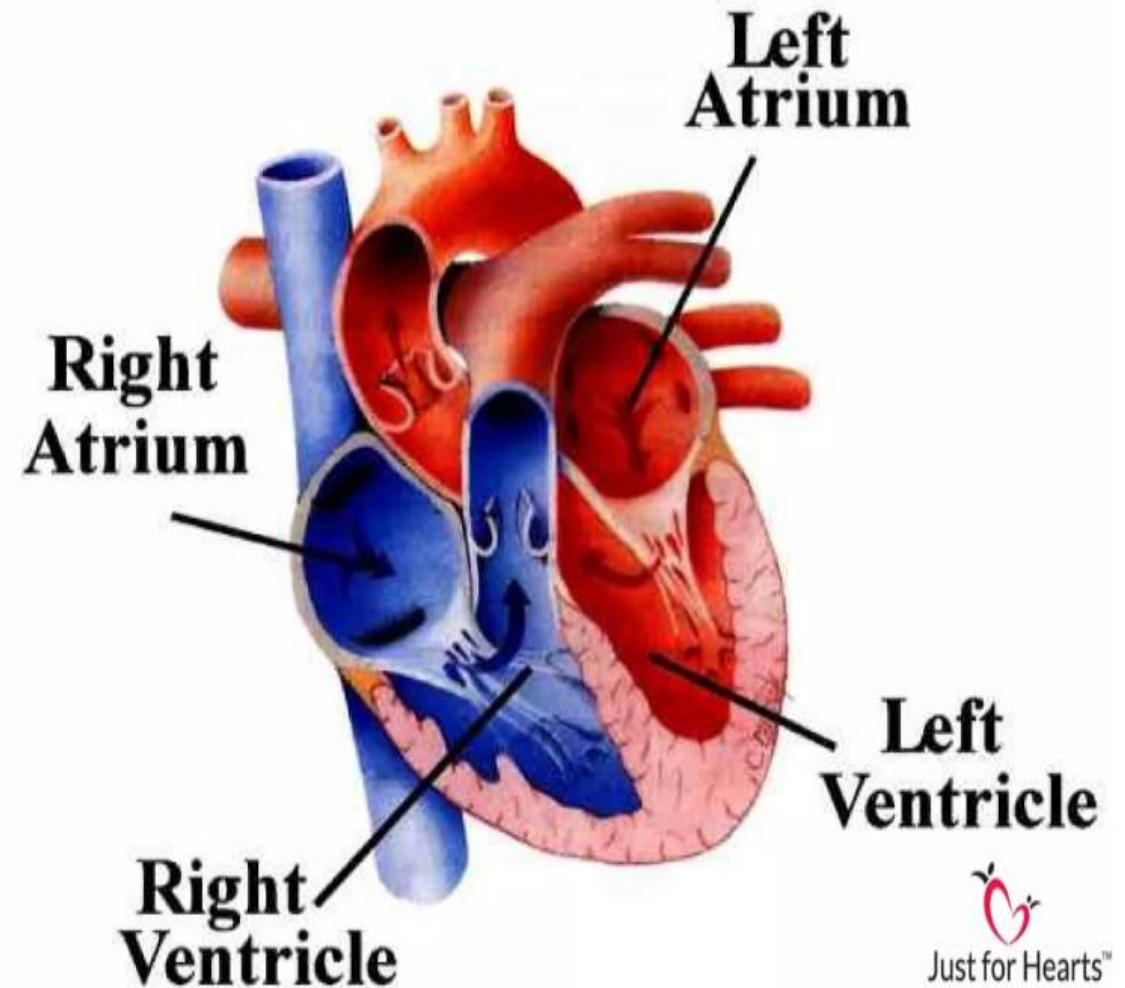


VALVES OF HEART

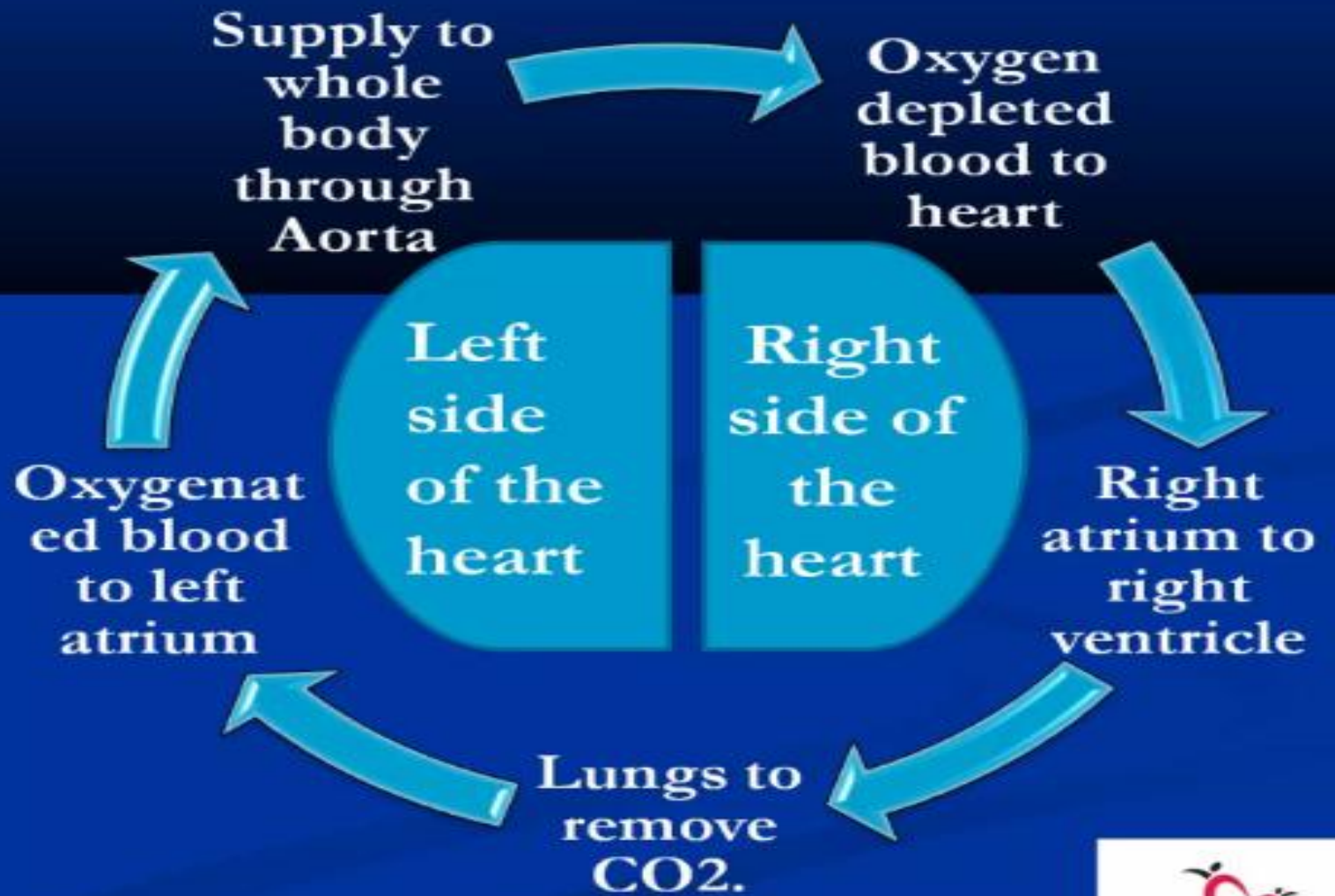
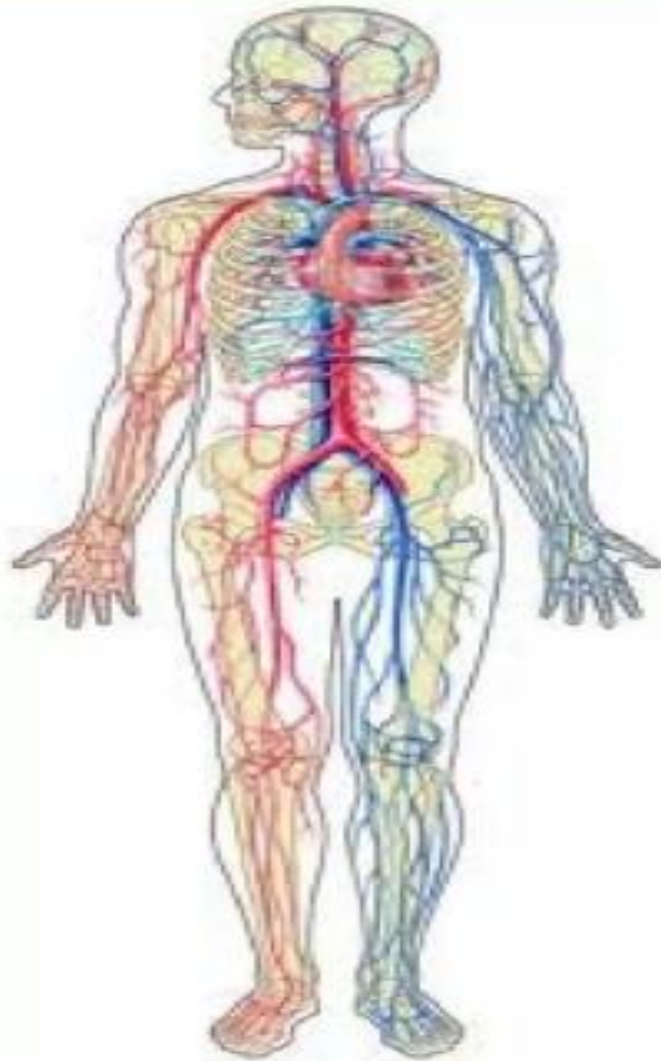


**“TO AVOID MIXING OF BLOOD
BETWEEN VENTRICLES AND ATRIA
VALVES ARE IMPORTANT”**

CHAMBERS OF HEARTS



How does blood flow through your heart?



Heart sounds

The heart sounds heard with a stethoscope are caused by vibrations originating in the heart and the major vessels. The opening and closing of the heart valves contribute greatly to the heart sounds; turbulent flow occurs at these times and the vibrations produced are often in the audible range.

The physics of some cardiovascular diseases

- Heart diseases often have physical component.
- Many of these diseases, for example, increase the work load of the heart or reduce its ability to work at a normal rate.
- $W = P \Delta V$ equation of work done
- $T = P R$ Laplace equation

- The work done by the heart is roughly the tension of the heart muscle times how long it acts. Anything that increase the work load of the heart. For example

- 1- Hypertension

Causes the muscle tension to increase in proportion to the pressure, due to Laplace law $T = PR$ so the high blood pressure causes to increase the work done by equation $W = P\Delta V$

- 2-Tachycardia

A fast heart rate increases the work load since the amount of time the heart muscle spend contracting increases.

- 3-Enlargement of the heart and reduction in the ability of the heart to provide adequate circulation.

- If the radius of the heart is doubled, the tension of the heart muscle must also be doubled if the same pressure is to be maintained.
- Since the heart muscle is stretched, it may not be able to produce sufficient force to maintain normal circulation, the stretched heart muscle is also much less efficient than normal heart muscle

- Congestive heart failure

- An enlargement of the heart and a reduction in the ability of the heart to provide adequate circulation.
- Law of Laplace : if the radius of the heart is doubled, the tension of the heart muscle must also be doubled.
 - If the same blood pressure is to be maintained.
 - Since the heart muscle is stretched, it may not be able to produce sufficient force to maintain normal circulation.
 - The stretched heart muscle is also much less efficient than normal heart.
 - It consumes much more O_2 for the same amount of work.
 - The treatment is to replace the heart surgically

Heart valve defects are of two types: -

- Stenosis: - the valve does not open wide enough. In stenosis the work of the heart is increased because a large amount of work is done against the obstruction of the narrow opening, and the blood supply to the general circulation is reduced.
- Insufficiency: - the valve does not close well enough. In insufficiency some of the pumped blood flows back into the heart so that the volume of the circulated blood is reduced.
- The treatment for both type of defects can be replaced by artificial valves or be replaced by cadaver heart valves

Blood vessels diseases

- Aneurysm is a weakening in the wall of artery
 - Resulting in a diameter increase ,increases tension in the wall of arteriols
 - The wall will blow out, if an aneurism does not rupture it is often fatal – if rupture is in the brain (cardiovascular accident)
- Vessel problem is the formation of sclerotic plaques on the walls of artery
 - Turbulent flow and produce a notable murmur
 - The narrowing of the artery will cause an increase of the Bernoulli effect.
 - The plaque may dislodge and travel with the blood supply to the affected part
 - In the brain ,it will produce a stroke (CVA)

<https://medicine.uodiyala.edu.iq/wp-content/uploads/2022/10/physics-cardio.pdf>

Thank You...